

ROKESHUVARAJ N

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ACADEMIC QUALIFICATION

Master of Science (Aerospace), Virginia Polytechnic Institute and State University, Blacksburg, (Pursuing CGPA-3.9)
Bachelor of Technology, (Aerospace), Alliance College of Engineering & Design (ACED), Affiliated to Alliance University, Bengaluru, C.G.P.A: 3.1/4.0; August 2020

PROFESSIONAL EXPERIENCE

Software Engineer, Avohi Info Tech, Bengaluru, February 2021 - Nov 2024

Avohi Info Tech offers web development, software services, and digital marketing solutions for businesses

- Built reusable integration components, optimized data flows, and implemented effective error-handling mechanisms like automated exception tracking and logging systems
- Developed the Frond-end of a food delivery application

INTERNSHIP

Aeronautical Intern, Kaashiv Infotech, Chennai, June-July 2019

Kaashiv Infotech offers internships, and project solutions to equip students and professionals with technology knowledge

- Designed a rocket used as a model and for analyzing which the most suitable material for a rocket was decided
- Assisted in organizing and analyzing relevant data to support investigations of ground damage incidents of operational aircraft and employee injuries
- Provided support to ground safety programs and company accident/incident and injury reduction efforts by proactively developing safety solutions like improved safety awareness training programs which led to a 30% increase in safety awareness among employees

ACADEMIC PROJECTS

Title: Design and Development of Wireless Power Transmission System for UAV

Duration: February-May 2020

Team Size: 4

Summary: Designed and developed a wireless power transmission system for Unmanned Aerial Vehicles (UAVs) to extend their flight time without landing. Laser light was used to refuel the drone while it remained airborne, ensuring a continuous power supply. Researched and selected laser technology for drone charging, ensuring efficient power transmission and maximizing energy transfer capabilities.

Title: Voice-Controlled Drone

Duration: September-November 2019

Team Size: 3

Summary: The project focused on the design of the first stage of an Unmanned Aerial Vehicle (UAV). In the final stage, an Arduino board with pre-programmed voice commands was used in combination with an Android application to control the drone. Developed and programmed voice command integration between the Arduino board and the Android application for user-friendly control of the UAV, improving the overall user experience.

Title: Balloon Satellite

Duration: March-May 2019

Team Size: 5

Summary: The project's initial phase involved thorough preparation to address weather obstacles for balloon satellites' orbital success. The satellite was designed and coded meticulously, aligning with specific requirements and mission goals. Effective thermal insulation was implemented to safeguard against extreme temperatures. Established communication systems and derived valuable insights from the collected data for informed decision-making.

Title: Aircraft Wing Fabrication

Duration: February-April 2018

Team Size: 3

Summary: The project aimed to fabricate miniature aircraft wings for model airplanes, with a focus on aerodynamics, structural integrity, and weight considerations. Collaborated on the design, material selection, shaping of the wing core, application of a protective covering, and integration of the wing. Various tools like CAD software, cutting tools, adhesives, protective coatings, and measurement instruments were utilized throughout the process to ensure accuracy and quality.

Title: Solid Propellant

Duration: Jan 2017 - April 2017

Team Size: 3

Summary: The objective of the project was to develop and test solid propellants using a combination of potassium nitrate and sugar for amateur rocketry. Researched propellant ratios, conducted experiments, designed the rocket motor, and analyzed the experimental data. To accomplish this, a combination of lab equipment like balance, beakers, etc., measurement instruments like Vernier Caliper, Stopwatch, and computer software like spreadsheets, simulation tools, etc. were utilized.

PUBLICATION AND PAPER PRESENTATION

- Author: Rakeshuvaraj N, Co-Authors: Vivek D, Cholasamudram Sai Pavan Reddy, Vijayalakshmi Bharathi K, 'Design and Fabrication of Balloon Satellite' published in the International Journal of Engineering Applied Sciences and Technology (IJEAST) Digital Library, Vol 4, Issue 9, ISSN No. 2455-2143, Page no 223-240, March 2020
- Presented a paper titled 'Balloon Satellite Design' at the International Conference on Emerging Trends in Engineering and Technology (ICETET) held at Paavai Engineering College, Namakkal, September 2019

INDUSTRIAL VISIT

Taneja Aerospace & Aviation Limited, Denkanikotta, September 2018

Learnt advanced aerospace technologies like composite materials and observed manufacturing processes of "Aerostructures"

ONLINE COURSES AND WORKSHOP

- Nanotechnology Course offered by Duke University, North Carolina State University, and the University of North Carolina at Chapel Hill through Coursera, November 2020
- Arduino Programming Course offered by the University of California, Irvine through Coursera, November 2020
- IoT and Embedded Systems offered by the University of California, Irvine Course through Coursera, November 2020
- Raspberry Pi Programming Course offered by the University of California, Irvine through Coursera, October 2020
- Search Advertising and Digital Advertising Course offered by the University of Colorado (CU) through Coursera, October 2020
- Artificial Intelligence (AI) Course offered by IBM through Coursera, October 2020
- How to Create a Website Course offered by the State University of New York (SUNY) through Coursera, October 2020
- VR & 360 Video Production Course offered by Google Daydream through Coursera, October 2020
- Initiating and Planning Projects Course offered by the University of California, Irvine through Coursera, August 2020
- 'Micro Aerial Vehicle' Workshop conducted by Jet Aerospace, held at Adhiyamaan College of Engineering, Hosur, February 2018

TECHNICAL SKILLS

- CAD/CAM Software: AutoCAD, CATIA, SolidWorks, ANSYS
- Modeling & Control Software: MATLAB, Simulink
- Programming Languages: C, C++, JAVA, Html & CSS
- Integrated Development Environment: Eclipse, Visual Studio
- Simulation & Mission Analysis: STK Software
- Programming Boards: Raspberry Pi & Arduino

ADDITIONAL ACTIVITIES AND ACHIEVEMENTS

- Stood first in Model Making out of 7 participants in the Aero-Mech Club Event held at ACED, Bengaluru, April 2019
- Volunteered to organize the departmental fest at ACED, Bengaluru, April 2018
- Basic STK Software Training, ACED, Bengaluru, April 2018
- Secured Grade -1 in Electronic Keyboard through Trinity College London, October 2010